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RESEARCH ARTICLE

Anti-cancer Effect of Cyanidin-3-glucoside from Mulberry *via* Caspase-3 Cleavage and DNA Fragmentation *in vitro* and *in vivo*

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Abstract: ***Background*:** Fruits of *Morus alba* L. (mulberry) have various bioactive compounds such as polyphenols and anthocyanins and used as a herbal medicine. However, the anti-cancer effects and molecular basis have not been elucidated.

***Methods*:** We isolated the cyanidin-3-glucoside in various cultivar of mulberry by acidified-methanol extraction methods. This molecule were compared mass spectroscopic properties by LC-MS/MS and analyzed by ^{1}H and ^{13}C NMR. We examined the anti-cancer effect with molecular mechanisms of the cyanidin-3-glucoside on MDA-MB-453 human breast cancer cells and xenograft animal model.

***Results*:** The treatment with the mulberry cyanidin-3-glucoside decreased cell viability in a dose-dependent manner with alteration of apoptotic protein contents, and DNA fragmentation, suggesting that cells undergo apoptosis. Supporting the observations, Treatment with the cyanidin-3-glucoside showed active apoptosis by caspase-3 cleavage and DNA fragmentation through Bcl-2 and Bax pathway. Indeed, cyanidin-3-glucoside inhibits tumor growth in MDA-MB-453 cells-inoculated nude mice. Tumor growth of xenograft nude mouse was significantly reduced compared to the control group by the cyanidin-3-glucoside**.**

***Conclusion*:** The data demonstrate that cyanidin-3-glucoside isolated from mulberry induced apoptosis in breast cancer (MDA-MB-453) cells, and therefore, has a potential as an anti-cancer agent. These results show that mulberry cyanidin-3-glucoside inhibit the proliferation and growth *in vitro* and *in vivo* model and, indicating the inhibition of tumor progression.

Keywords: Cyanidin-3-glucoside, *Morus alba* L., mulberry, anti-cancer, caspase-3, DNA fragmentation, nude mice.

1. INTRODUCTION

Morus alba L. fruits (mulberry) have been regarded as a healthy food and are widely consumed since ancient times in Asia countries including Korea, China, Serbia, and Japan. Studies have shown that administration of mulberry extracts markedly reduces oxidative stress [1, 2], improves lipid profile [3], and shows ameliorative effects in animal models of various diseases [4]. A standardized extract of mulberry contains relative high polyphenol content and displays similar results [5, 6]. Recent studies have demonstrated the importance of mulberry extracts as a dietary antioxidant in prevention of oxidative damage, decrease of cancer cell proliferation, and modulation of cell growth *in vitro* [7]. However, the molecular mechanisms involved in anticancer properties of mulberry cyanidin-3-glucoside remain unclear.

In this study, we evaluated anti-cancer effects of cyanidin-3-glucoside (C3G) isolated from mulberry using MDA-MB-453 cells, in an *in vitro* model of breast apocrine carcinomas. The molecular basis of anti-cancer effects, focusing on apoptosis cascade, was investigated. We also demonstrated that administration of mulberry C3G significantly reduces tumor sizes using *in vivo* model.

2. MATERIALS AND METHOD

2.1. Preparation of C3G from Mulberry

The frozen fruits of *Morus alba* L. (mulberry) for this study were purchased from Chonbuk National University-Buan Regional Innovation System (Jeonju, Korea). The Buan mulberry C3G was prepared with methanol containing 1% HCl overnight at room temperature. The partial C3G fraction was centrifuged at 2,500 rpm for 10 min and then filtered using 0.45 µm syringe filters. The filtrate was fractionated by liquid chromatography (Dongil Shimaz) with C18 column (YMC-Hydrosphere, C^{18} 5 µm, 4.6 × 150 mm) in preconfigurated condition (mobile phase: Pump A - 0.1% Acetic acid, Pump B - AcCN (A:B=95:5~5:95, 35min)). The flow rates were maintained with 1 ml/min and detection wavelength was 254 nm.

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Fractionated mulberry C3G (Fig. 2) were dried using a freezing drier and then kept in a -80°C freezer before use. The prepared mulberry C3G and pure C3G standard (Sigma-Aldrich, CA, USA) were identified using high-performance liquid chromatography (HPLC) (Waters, MA, USA) retention time and compared.

2.2. LC-MS/MS Analysis of Mulberry C3G

Agilent 6410B Triple Quadrupole LC/MS (Agilent Technologies, Wilmington, USA) equipped with an ESI source was employed for the analysis. Analysis was performed at the Center for University-wide Research Facilities in Chonbuk National University (Jeonju, Korea). Kuromanin chloride (Lot#BCBQ9084V) was purchased from Sigma Aldrich (St. Louis, USA) and used as reference standard. 100 mg of each sample was mixed with 900 µl of methanol and centrifuged. Aliquots of 5 µl of the processed samples were injected into the HPLC system (1200 Series LC, Agilent Technologies, Wilmington, USA) fitted with Phenomenex Synergi Hydro-RP 4 µm 80 Å 150 x 2 mm column, maintained at 30°C. ESI was operating at +3,000V and a source temperature of 380°C. Capillary voltage, cone voltage and source offset were set at 3kV, 30kV and 30V respectively. The gas flow of desolvation and the cone was set at 650 l/hr, 150 l/hr with a nebulizer pressure of 15 bar. A mobile phase composed of 0.1% formic acid in distilled water (Buffer A) and 0.1% formic acid in acetonitrile (Buffer B) was used to separate the analysts and pumped into the ESI chamber at a flow rate of 0.2ml/min for 20 min. Fragmentor voltage and collision voltage was set at 70V. Detection of the ions was carried out in the multiple-reaction monitoring mode (MRM), by monitoring the transition pairs of m/z 450.1 → 288. Data acquisition was performed with the MassHunter Software (Version B.04.00).

2.3. Identification of Mulberry C3G

Microanalysis for ^{1}H and ^{13}C NMR (nuclear magnetic resonance) and DEPT (distortionless enhancement by polarization transfer) analysis of mulberry extracts in DMSO-d_6 were measured using a JEOL JNM-ECA600 600 MHz FT-NMR spectrometer (JEOL, Tokyo, Japan). Analysis was performed at the Center for University-Wide Research Facilities in Chonbuk National University (Jeonju, Korea).

2.4. Animals

Five-week-old female Crl:NU-*Foxn1*^{nu} mice were bought from Orient Bio (Sungnam, Korea). Mice were housed under 12 hr light/12 hr dark conditions and allowed free access to food and tap water. Bedding was changed once a week, and the temperature and humidity were controlled. All animal studies were performed according to a protocol approved by the Institutional Animal Care and Use Committee of the Nambu University (Approval No: 201408). Mice were acclimatized for one week and divided randomly into three groups.

2.5. Cell Culture

A human breast cancer cell line, MDA-MB-453 was purchased from American Type Culture Collection (ATCC, VA, USA). Cells were grown in Dulbecco's Modified Eagle's Medium (DMEM) supplemented with 10% fetal bovine serum (FBS) at 37°C in a humidified atmosphere with 5% CO_2 and 95% air condition to ~80% confluence. A subculture was performed every 2-3 days.

2.6. Determination of Cell Viability

The 3-(4,5-dimethylthizaol-2-yl)-2,5-diphenyl tetrazolium bromide (MTT) assay was used to determine cell viabilities [8]. Briefly, MDA-MB-453 cells were seeded onto 96-well culture plates at a density of 5×10^3 cells per well. The cells were treated with mulberry C3G for 48 hr to 72 hr. Then, 100 µl of MTT solution (1 mg/ml) was added to each well.

The resulting crystals were dissolved in dimethyl sulfoxide (DMSO). The formation of formazan was measured by reading absorbance at a wavelength of 570 nm using microwell plate reader (Molecular Devices, CA, USA).

2.7. Western Blotting

Cells were washed twice with phosphate-buffered saline (PBS), and solubilized with cell lysis buffer (1% Triton X-100, 150 mM NaCl, 10 mM Tris, pH 7.5) supplemented with complete protease inhibitors (100_M PMSF, 2 mM $NaVO_4$, and 5 mM NaF; Sigma-Aldrich, MA, USA), and centrifuged at 15,000 rpm at 4°C for 20 min. Protein concentration was measured by the bicinchoninic acid (BCA) protein assay (Pierce, Rockford, IL, USA) using bovine serum albumin (BSA) as the standard. The proteins were separated on sodium dodesyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE) and gels were transferred to polyvinylidene fluoride (PVDF) membranes (Bio-Rad Laboratories, Inc., Hercules, CA, USA). The membranes were blocked with 3% IgG-free bovine serum albumin (Invitrogen, MA, USA) in PBS containing 0.05% Tween 20 (PBS-T) at room temperature for 1 hr and then incubated with specific primary antibodies in 3% BSA-PBS-T for overnight at 4oC. After three time washes with PBS-T, the blots were incubated with horse radish peroxidase (HRP)-conjugated secondary antibodies (1:1,000, Santa Cruz, TX, USA) at room temperature for one hr, and proteins were visualized with the ECL substrate kit (Amersham Biosciences, CA, USA). The visualized images were captured with a software (ImageQuant LAS-4000, GE healthcare, Uppsala, Sweden) and and analyzed Image J (NIH, USA).

2.8. DNA Fragmentation Analysis

For DNA fragmentation analysis, MDA-MB-453 cells were washed twice with PBS by centrifugation at 3,000×g for 10 min, and resuspended with lysis buffer (10 mM Tris-HCl, pH 7.4, 10 mM EDTA, 0.5% Triton X-100). The lysates were centrifuged at 16,000×g for 20 min, and the resulting supernatants were treated with proteinase K (final concentration of 100 µg/ml, Boehringer Mannheim, GmBH) and DNase-free RNase (final concentration of 100 µg/ml, Boehringer Mannheim, GmBH) at 37°C for 1 hr. Ethanol precipitate of genomic DNA was recovered by centrifugation centrifuged at 16,000×g for 20 min. Genomic DNA was dissolved with Tris-EDTA buffer and electrophoresed on 2% agarose gel. The separated DNA was stained with ethidium bromide (5 mg/ml) and visualized under UV light and obtained images (LAS-1000, Fujifilm, Tokyo, Japan).

2.9. Tumor Transplanted Mouse Model

Animals were administered 17β-estradiol (Sigma-Aldrich, CA, USA) with ALZET minipump (2 mg/mice, release rate: 1,000 µg/day, for 28 day, dissolved in 10% EtOH-50% glycerol-DW)(Sigma-Aldrich, CA, USA) in accordance with manufacturer's instructions. At week two, MDA-MB-453 cells (2×10^5 per 0.1 mL of PBS) were injected subcutaneously into mice (Day 0), and tumor growth was monitored by measuring the tumor sizes with calipers (tumor size = length × width) during experimental period. The mulberry C3G was orally administered daily two time at a concentration of 1.6 and 3.2 g/kg. For the vehicle treatment group, the same volume of distilled water was administered.

To calculate T/C value, we used priviously reported formular in below [9].

$$T/C\% = [(RVtreated)/(RVcontrol)] \times 100$$ were collected.

2.10. Statistics

Data are expressed as mean ± S.D. Significant differences between groups were determined using the unpaired Student's t test. Statistical significance was set at $p<0.05$.

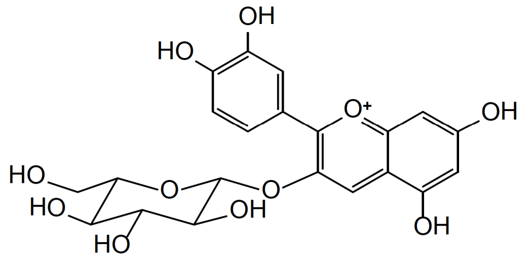

Fig. (1). Structure of Cyanidin 3-O-glucoside chloride (C3G, Kuromanin, Asterin, Chrysanthemin; CAS Number 7084-24-4; C_{21} H_{21} ClO_{11}; MW 484.84).

3. RESULTS

3.1. Isolation and Characterization of Mulberry C3G

To observe whether mulberry C3G (like commercial C3G) (Fig. **1**) affects viability of MDA-MB-453 cells, we performed MTT assay. Prior to the MTT assay, we analyzed two mulberry cultivars ‘Chungil’ and ‘Iksu’, to compare contents of C3G. In general, both cultivar were well grown in many provinces of South Korea, and we observed high contents of C3G in ‘Chungil’ (Fig. **2**). Hence, we choose this cultivar our study. Futhermore, we characterized mulberry C3G by LC-MS/MS analysis. In this study, kuromanin chloride (Sigma-Aldrich, CA, USA) was chosen as standard C3G compound. Mulberry extract was dissolved in 50% methanol. Diluted sample (1:50) with A buffer (0.1% FA in DDW) was applied to LC-MS/MS device. The standard mixture was applied as 1, 10, 100 and 1,000 ppb (1, 10, 100, and 1,000 µg/l, respectively). Comparison of retention time, which correspond to masses of the predicts, of mulberry C3G and standard compounds revealed the relative abundance of C3G in the extract as shown in Fig. **3**. C3G is the most abundant compounds in concentrates from HPLC fractions of mulberry acidified methanol (Fig. **2**). In this study, C3G was found as 88.35 ppm in 0.1 mg/ml (w/v) of mulberry C3G in 50% methanol. The charaterization of C3G by LC-MS/MS suggests that fractionated mulberry extracts has a most amount with C3G present in the fractionated sample according to HPLC analysis. In addition, the complete structures of isolated mulberry extract was established by ^{1}H-NMR and ^{13}C-NMR spectra analysis as follow: ^{1}H-NMR (600 MHz, DMSO-d_6) δ 12.28 (bs, OH), 8.89 (s, $1H_{4}$), 8.23 (dd, $1H_{6¢}$, J = 8.4, 2.4 Hz), 8.02 (d, $1H_{2¢}$, J = 2.4 Hz), 7.06 (d, $1H_{5¢}$, 8.4 Hz), 6.97 (bs, $1H_{8}$), 6.82 (d, $1H_{6}$, J = 2.4 Hz), 5.36 (d, $1H_{1''}$, J = 7.8 Hz), 3.68 (m, $1H_{6b''}$), 3.45 (m, $3H_{2''5''6a''}$), 3.33 (t, $1H_{3''}$, J = 9.0 Hz), 3.19 (t, $1H_{4''}$, J = 9.0 Hz); ^{13}C-NMR (150 MHz, DMSO-d_6) δ 168.11, 161.06, 157.28, 155.37, 154.01, 145.87, 144.77, 134.28, 126.46, 116.43, 116.12, 111.70, 102.04, 101.49, 93.68, 77.22, 76.11, 72.73, 69.23, 60.34. A ^{13}C NMR DEPT(135 deg) experiment demonstrated that the carbon skeleton matched with the cyanidin 3-O-β-D glucoside structure as follow: eleven methine carbons (δ 134.28, 126.46, 116.43, 116.12, 102.04, 101.49, 93.68, 77.22, 76.11, 72.73, 69.23) and one methylene carbon (δ 60.34). The small amount of α–glycoside isomer was also detected by ^{1}H- NMR spectra, which might be formed during extracting with 1% HCl in methanol. Generally, for α–anomer, coupling constants of $J_{1''2''}$ = 2−5 Hz were observed, while $J_{1''2''}$ = 6−10 Hz were characteristic for the β–anomer. Additionally, the axial anomeric proton (1"-H) chemical shift of β–anomer always resonates at a higher field than the equatorial proton of the corresponding α–anomer[10]. Therefore, the fractionated mulberry extracts having 5.65 ppm ($J_{1''2''}$ = 1.8 Hz) and 5.36 ppm ($J_{1''2''}$ = 7.8 Hz) which were assigned from their ^{1}H-NMR

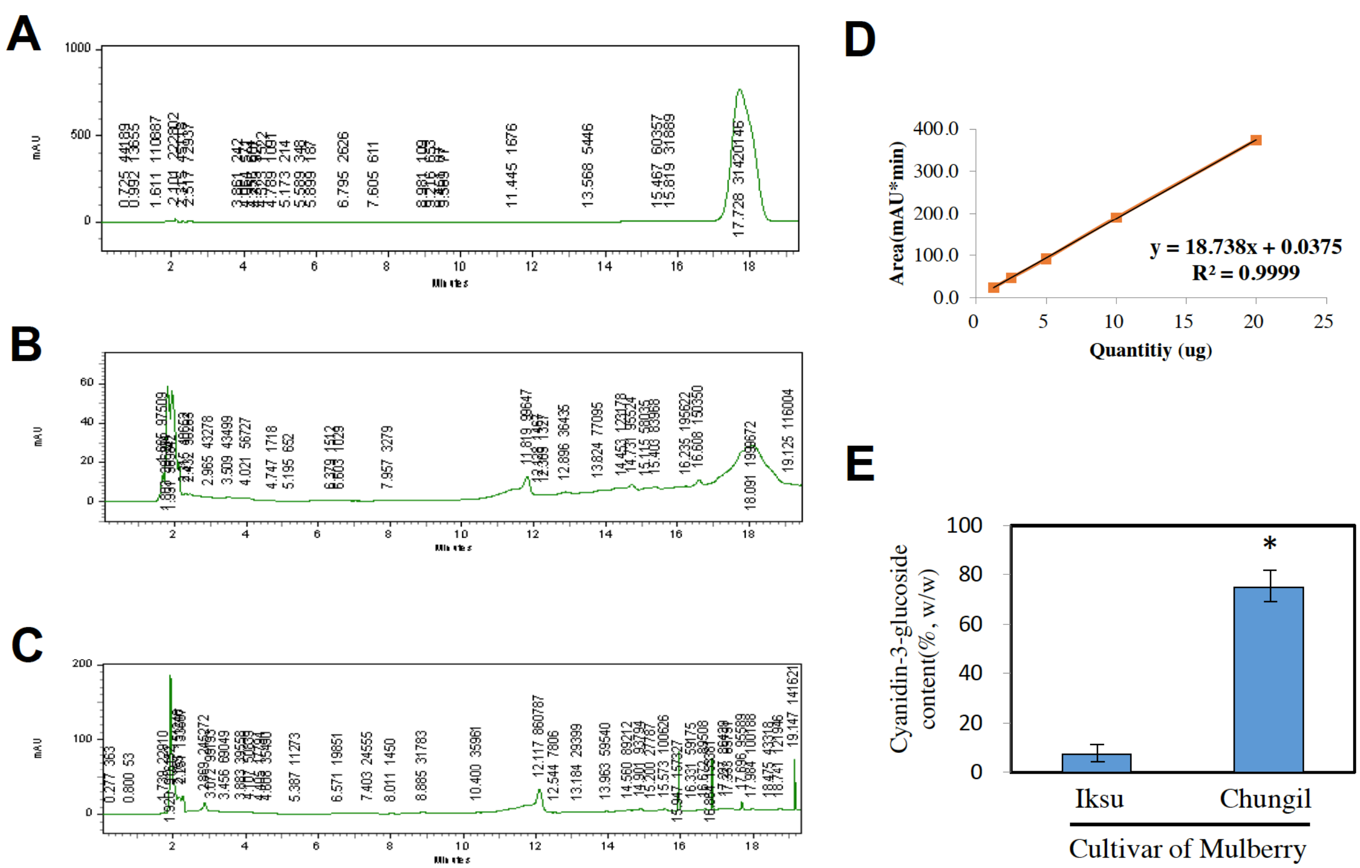

Fig. (2). Direct comparison of HPLC analysis profiles between two cultivar of the *Morus alba* L. (**A**) Standard compound of C3G, (**B**) Mulberry C3G with cultivar “Iksu” and (**C**) “Chungil”. (**D**) Standard curve of C3G and (**E**) comparison of C3G content between both cultivar.

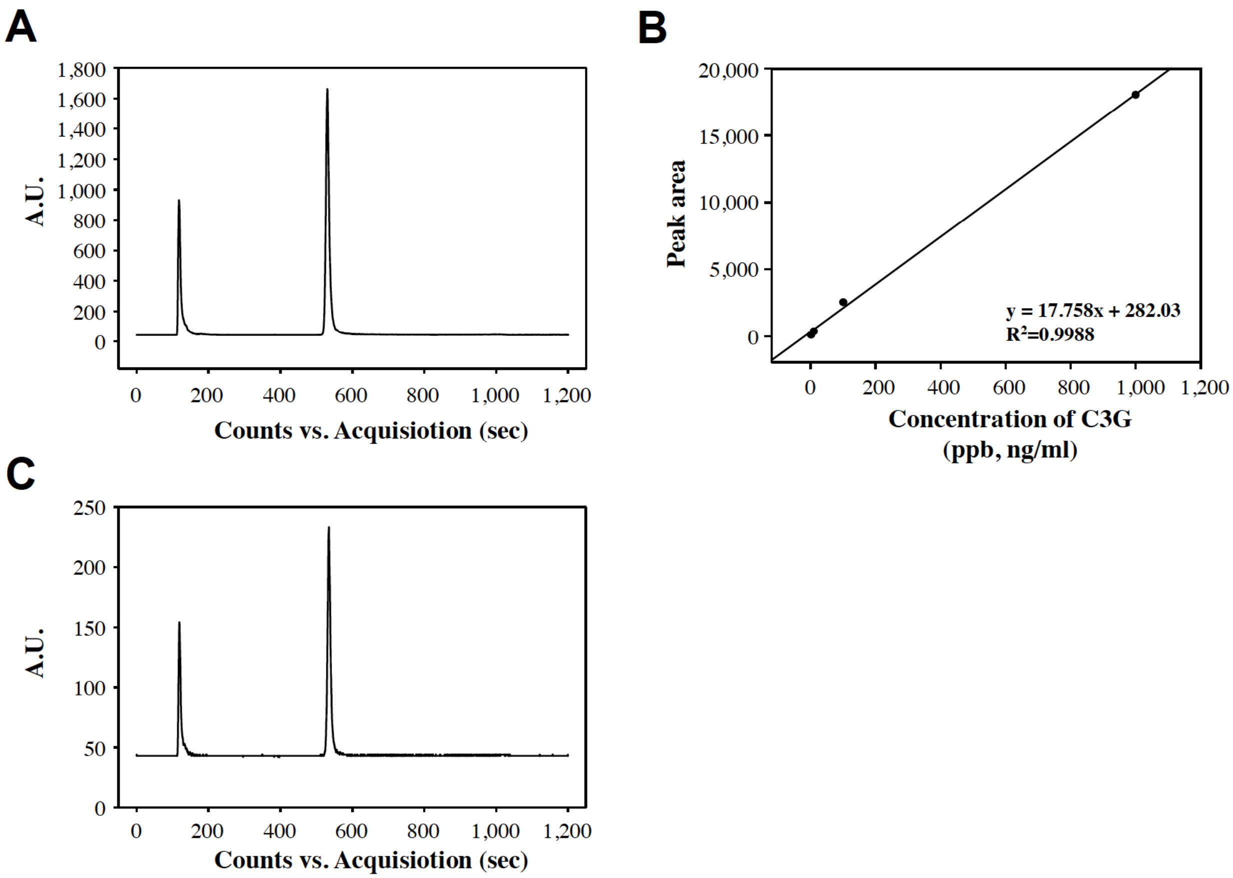

Fig. (3). LC-MS/MS chromatography of (**A**) 1,000 ppb standard mixture and (**B**) standard curve of standard molecule with 1, 10, 100, and 1,000 ppb solution of kuromanin chloride. (**C**) Mulberry C3G fraction (0.1 mg/ml (w/v) in 50% methanol).

spectra were found to be α–and β–anomer, respectively. Morever, we concluded that the major material of isolated mulberry extracts was identified as the cyanidin 3-O-β-D glucoside by comparing the spectra with the literature [11].

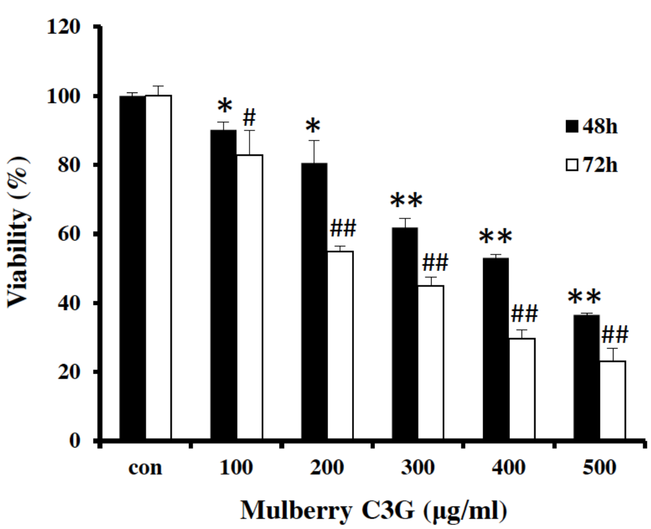

Fig. (4). Time and dose-dependent increases in cytotoxicity. Mulberry C3G caused appreciable cytotoxicity in MDA-MB-453 cells in CO_2 incubator at 48 and 72 h time points. Data are expressed Mean ± S.D for statistics. *P < 0.05 and **P<0.01 versus control group. #P<0.05 and ##P<0.005 versus control group.

3.2. Cytotoxic Effects of Mulberry C3G on MDA-MB-453 Cells

Mulberry C3G-induced cyatotoxicity in MDA-MB-453 cells tested with various time and dose conditions. We observed that treatment with mulberry C3G increased cell death of MDA-MB-453 cells in a time and dose-dependent manner (Fig. **4**). At a concentraion of 200 μg/ml at 72 hr, mulberry C3G enhanced the cytotoxicity than at 48 hr in this cell line.

3.3. Mulberry C3G Induces Apoptosis in MDA-MB-453 Cells

To examine whether cytotoxic activity of mulberry C3G is related with cell apoptosis, the level of anti- and pro- apoptotic proteins was determined by Western blotting (Fig. **5**). The results from Western blot analysis showed that treatment with mulberry C3G increased the level of cleaved caspase-3 per caspase-3, stastiscally in a dose dependent manner (Fig. **5A**), while the level of caspase-3 and Bax protein was not significantly changed between control and mulberry C3G-treated groups. In contrast, the level of Bcl-2 was significantly reduced after treatment with the mulberry C3G compared to the level of control group (Fig. **5B**). These results suggest that mulberry C3G induced activation of caspase-3 and down-regulate the expression of Bcl-2.

3.4. Mulberry C3G Causes DNA Fragmentation

Apoptosis is a form of programmed cell death characterized by various phenomena in cells leading to nuclear DNA breakdown into multiples of small size of oligonucleosomal fragments. To further evaluate the mulberry C3G-induced apoptosis, MDA-MB-453 cells were treated with various concentrations of mulberry C3G and then oligonucleosomal laddering was determined by agarose gel electro-

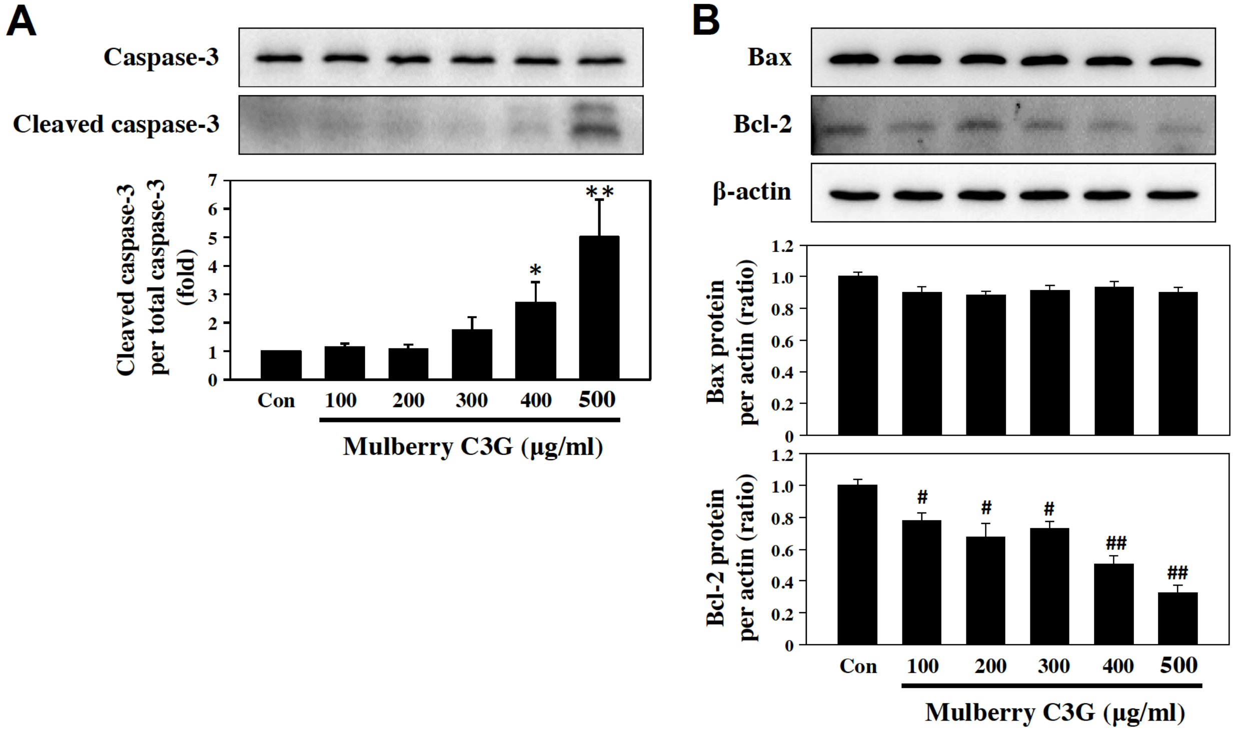

Fig. (5). Mulberry C3G increases cleaved caspase-3 per caspase-3 and reverses Bcl-2 protein expression in the MDA-MB-453 cells. Western blot showing images of (**A**) cleaved caspase-3 per caspase-3 and (**B**) Bax, and Bcl-2 protein expression per actin. Cells (1×10^6 cells/ml) treated with indicated concentration of mulberry C3G in CO_2 incubator for 72 h treatment. *P < 0.01 **P<0.001, #P<0.05, and ##P<0.005 versus control group.

phoresis (Fig. **6**). Treatment with mulberry C3G showed active apoptosis in a concentration-dependent manner, showing an increase in small size of oligonucleosomal fragments.

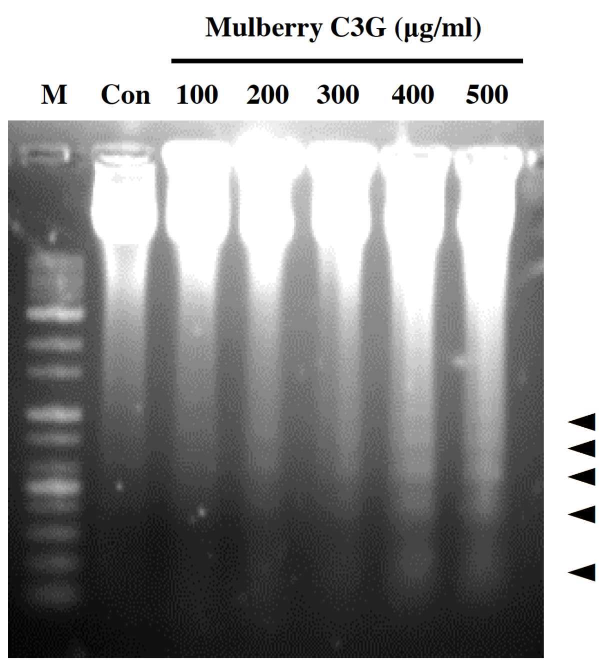

Fig. (6). DNA fragmentation in MDA-MB-453 cells treated with mulberry C3G. DNA extracted from cells as described under the Materials and Methods. Intranucleosomal DNA fragmentation was visualized by staining with ethidium bromide after agarose gel electrophoresis. Cells (1×10^6 cells/ml) were treated with various concentration of mulberry C3G in CO_2 incubator for 72 h. Lane m, DNA molecular marker. Arrows indicates fragmented DNA.

3.5. Mulberry C3G Exhibits Anti-tumor Activity *in vivo*

When tumor-transplanted nude mice were fed with mulberry C3G two times per day for 25 days reduced tumor size compared to that of control mice (Fig. **7**). Enlarged tumor size in control group was diminished after mulberry C3G treatment. At day 10, T/C% were 45.29 (1.6 g/kg) and 33.10 (3.2 g/kg). Then, T/C% were changed to 74.56 (1.6 g/kg) and 54.52 (3.2 g/kg). These anti-tumorigenic ability of mulberry C3G was a dose-dependent manner. In addition, there was no observed body weight loss for the treated group in comparison with the placebo treated group.

4. DISCUSSION

To date, natural compounds and their analogues have been shown to inhibit or reduce tumorogenesis. Phytochemicals as cancer preventives have received great attention for suppressing cancer growth *via* various mechanisms [12, 13].

In the present study, we evaluated effects of mulberry C3G, on apoptosis and apoptotic protein markers using MDA-MB-453 cells. Our results revealed that treatment of the cells with mulberry C3G induces apoptosis and DNA fragmentation and increases the level of cleaved caspase-3 while decreasing Bcl-2. Supporting the observations, tumor growth in mice was significantly inhibited by the administration of mulberry C3G. These findings indicate that mulberry C3G have proven to prevent cancer growth *via* apoptosis signaling.

Apoptosis is a complicated process, which involves various signaling molecules such as caspases and Bcl-2. Varous caspases including caspase-6 and 9 are member of the cysteine-aspartic proteases family, which play a crucial role in apoptotic pathways [14-16] and chosen chemotherapeutic agents as caspase activators [14]. Among capases, caspase-3 is a cysteine peptidase involved in the activation of apoptosis cascade. Especially, caspase-3 cleavage is central executioner [17]. Bcl-2 regulates apoptotic processes

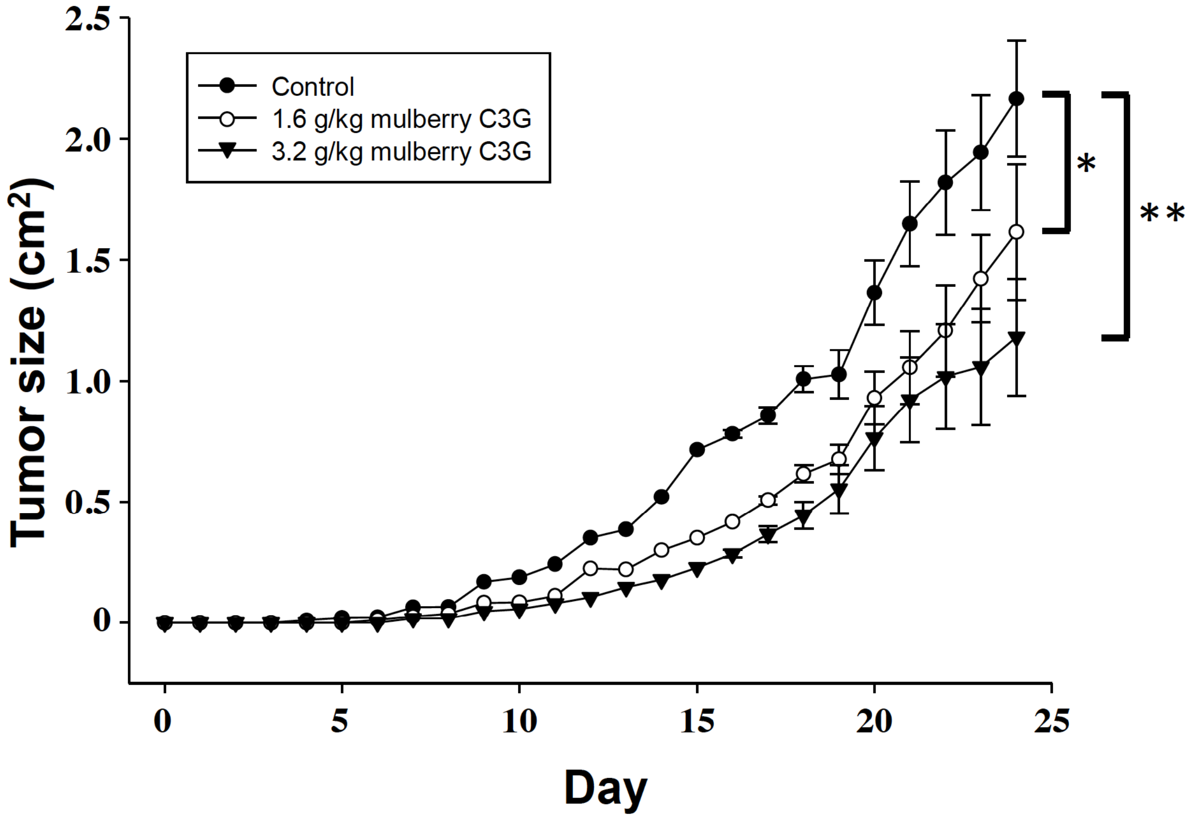

Fig. (7). Mulberry C3G feeding reduced the tumor growth in nude mice injected with MDA-MB-453 cells. Treated with mulberry C3G after the subcutaneous inoculation of the MDA-MB-453 cells to nude mice and the tumor growth curves recorded. *$P < 0.05$ and **$P<0.005$ versus control group.

through mutual neutralization of the bound pro- and anti-apoptotic proteins by homo/heterodimerization [18, 19]. In the present study, the results showed that mulberry C3G induces caspase-3 cleavage and reduces Bcl-2 level in MDA-MB-453 cells. Moreover, other caspases signaling pathway should be investigated in further studies. These observations suggest that mulberry C3G regulates apoptotic process *via* caspase-3 and Bcl-2 in MDA-MB-453 cells. [20, 21]. Furthermore, mulberry (*Morus alba* L. fruits) can orally be repetitively consumed. However, oral administration of mulberry does not cause acute and/or subchronic toxicity in all sex in rat model [16]. Recently, using other parts of *Morus alba* L. such as leaves [22-24], roots [25], and root bark [26, 27] are reported to contain effective molecules such as hydroxycinnamic acid esters, flavonol glycosides, and anthocyanins [5, 6, 28]. In our study, we used isolated C3G from one cultivar, Chungil, which contains high C3G and analyzed mass spectroscopic properties of mulberry C3G by LC-MS/MS methods. Moreover, we compared microanalysis data by ^{1}H and ^{13}C NMR of mulberry extracts with those of the literature [9]. And then, we elucidated that C3G are presented in mulberry extracts as major effective functional molecule. These molecule showed a significant role in reducing breast cancer cells and tumor size of cancer transplanted mice model. Although C3G has been known to show anti-cancer activity in an even higher dose [29], other compounds in mulberry can also exhibit some anti-cancer activity. Therefore, further studies are warranted to various cancer cells and normal cells like primary dermal fibroblast (data not shown), blood cells, and skin [30] of human sources, and mechanism-based experiments with agonist/antagonist and/or novel synthetic/metabolic analoges of C3G.

In conclusion, mulberry C3G significantly elevated caspase-3 cleavage *via* Bcl-2 process and DNA fragmentation in MDA-MB-453 cells. Mulberry C3G may act as an anti-cancer agent by activating apoptosis process in mice with breast cancer.

CONCLUSION

Mulberry C3G regulates apoptosis in MDA-MB-453 cells and may act as an anti-cancer agent by activating apoptosis process in rat with breast cancer.

ETHICS APPROVAL AND CONSENT TO PARTICIPATE

Not applicable.

HUMAN AND ANIMAL RIGHTS

No Humans were used for studies that are base of this research.

CONSENT FOR PUBLICATION

Not applicable.

CONFLICT OF INTEREST

The authors declare no conflict of interest, financial or otherwise.

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